Chapter 8b: Integral Equations and Computational Schemes

Applications of Monotone Operator Theory

Pathak, R. P.

An Introduction to Nonlinear Analysis and Fixed Point Theory

Outline

Integral Equations in Applied Mathematics

- Many physical phenomena modeled by integral equations
- Classical types:
  - Fredholm equations
  - Volterra equations
  - Hammerstein equations
  - Urysohn equations
- Monotone operator theory provides unified framework
- Key benefit: existence and uniqueness without linearization

Chapter 8 Focus

Applications of monotone operator theory to integral equations and computational schemes for finding approximate solutions.

The Hammerstein Operator

Definition (Hammerstein Operator Equation)

Let $\mathcal{H}$ be a Hilbert space. Given:

- $N_f : \mathcal{H} \to \mathcal{H}$ hemicontinuous and monotone
- $K : \mathcal{H} \to \mathcal{H}$ continuous and strongly monotone

Consider the equation:

$$(Kx, x) \geq c\|x\|^2 \quad \text{for some } c > 0$$

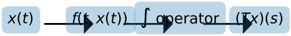
Theorem (Theorem 8.8)

Under the above conditions, equation (8.31) has a unique solution x that depends continuously on y and satisfies:

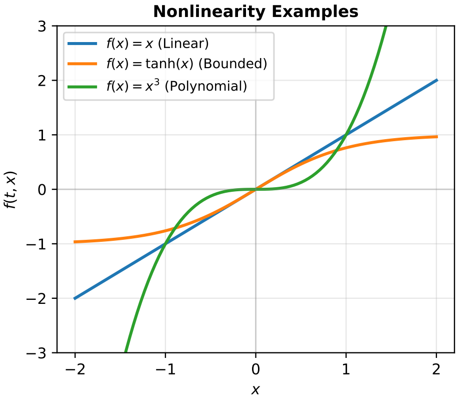
$$\|x\| \leq \frac{1}{c}\|K\|\|y\|$$

Structure of Hammerstein Equations

Hammerstein Operator: $(Tx)(s) = \int_{\Omega} k(s,t) f(t, x(t)) dt$



Properties:
• Nonlinear via $f(t, x(t))$\n• Integral via kernel $k(s,t)$\n• Applications: integral eqs.



Left: Block diagram showing composition of linear integral operator and nonlinear Nemytskii operator. *Right:* Examples of nonlinearities.

Concrete Hammerstein Integral Equation

Example (Example 8.6)

Consider the Hammerstein integral equation:

$$x(s) + \int_{\Omega} k(s,t) f(t, x(t)) dt = 0, \quad (8.34)$$

where K is the integral operator:

$$[Kx](s) = \int_{\Omega} k(s,t) x(t) dt$$

Key assumptions for existence:

1. $k(s,t)$ is symmetric, positive semidefinite, Hilbert–Schmidt kernel
2. $f(s,x)$ is monotone increasing in x with growth condition
3. There exist constants a, b, M such that:

$$f(s,x) > a + bx \text{ for } |x| > M$$

Angle Boundedness and Properties

Definition (Angle Bounded Operator)

Let X be a Banach space with dual X^* and $K : X \to X^*$ be a bounded linear monotone mapping. K is *angle bounded* with constant $c \geq 0$ if for all $x_1, x_2 \in X$:

$$|(Kx_1, x_2) - (Kx_2, x_1)| \leq 2c[(Kx_1, x_1)(Kx_2, x_2)]^{1/2}$$

Theorem (Theorem 8.10: Angle Bounded Case)

Let $K : X \to X^*$ be monotone, angle-bounded with constant $x \geq 0$. Let N_f be hemicontinuous with $(x_1 - x_2, N_f x_1 - N_f x_2) \geq -k\|x_1 - x_2\|^2$.
If $k(1 + c^2)\|K\| < 1$, then $x + KN_f x = 0$ has a unique solution with estimate:

$$\|x\| \leq \frac{\|K\|}{1 - (1 + c^2)k\|K\|}$$

Problem (Mixed Type Integral Equation)

Consider nonlinear integral equations of mixed type:

$$u(s) + \int_\Omega K_1(s,t) f_1(t, u(t))dt + \int_\Omega K_2(s,t) f_2(t, u(t))dt = 0 \quad (8.44)$$

where K_1, K_2 are kernel functions and f_1, f_2 are real-valued functions.

Operator Form: Let K_j, N_j be defined as before. The equation becomes:

$$u + K_1 N_1 u + K_2 N_2 u = 0 \quad (8.45)$$

General Case: For n Hammerstein operators:

$$u + \sum_{j=1}^{n} K_j N_j u = 0 \quad (8.46)$$

We work in product spaces $\mathcal{X} = X \times X \times \cdots \times X$.

Product Space Framework

Definition (Product Banach Space)

Define $\mathcal{X} = X \times X \times \cdots \times X$ (n copies) with norm:

$$\|U\| = \sqrt{\sum_{j=1}^{n} \|u_j\|^2}$$

where $U = [u_1, u_2, \ldots, u_n] \in \mathcal{X}$.

Define linear operator $\mathcal{K}$ and nonlinear operator $\mathcal{N}$ by:

$$\mathcal{K}(U) = [K_1 u_1, \ldots, K_n u_n] \in \mathcal{X}^*$$

$$\mathcal{N}(U) = [N_1 u_1, \ldots, N_n u_n] \in \mathcal{X}^*$$

Key Advantage: Conditions on individual operators translate to conditions on product space operators.

Generalized Hammerstein Equation

Definition (Generalized Hammerstein Type)

A nonlinear integral equation of generalized Hammerstein type is:

$$u(s) + \int_{\Omega} k(s,t,u) f(t,u(t))dt = v(s), \quad (8.52)$$

where the kernel function $k(s,t,u)$ depends on the solution u.

In operator-theoretic terms:

$$u + K(u) N_f u = v, \quad (8.53)$$

where $K(u) : X \to X^*$ is linear in second variable for fixed u.

Theorem (Theorem 8.17)

Let X be a real reflexive Banach space. For each $u \in X$, let $K(u) : X^* \to X$ be linear, angle-bounded with constant $c \geq 0$.
If $\|K(u)\| \leq r$ and $(u - u_2, N_f u_1 - N_f u_2) \geq -k\|u_1 - u_2\|^2$ with $kr < (1 + c^2)$, then $u + K(u)N_f u = 0$ has at least one solution.

Backwinkel-Schillings Approach

[Fixed Point Method] For equation $u + K(u)N_f u = v$, the Backwinkel-Schillings method:

1. For fixed $u \in X$, equation $u + K(u)N_f u = v$ is a Hammerstein equation in u
2. Define mapping $T : X \to X$ implicitly:

$$Tu + K(u)N_f Tu = v$$

3. Show that T has a fixed point
4. Study convergence of fixed point iterations

Advantage: Reduces to solving standard Hammerstein equations iteratively.

Urysohn's Integral Equation

Definition (Urysohn Equation)

Urysohn's integral equation is of the form:

$$u(s) + \int_{\Omega} K(s,t,u(t))dt = 0, \quad (8.58)$$

where Ω is a measurable subset of $\mathbb{R}^n$ with σ-finite measure, and $K(s,t,u)$ depends on all three variables.

Applications: Boundary value problems, radiative transfer, fluid mechanics

[Generality] The Urysohn equation is more general than Hammerstein:

- Hammerstein: $K(s,t)f(t,x(t))$ (separable)
- Urysohn: $K(s,t,u(t))$ (nonseparable in general)

Example (ODE to Urysohn Equation)

Consider boundary value problem:

$$\frac{d}{dt}\left(a(t)\frac{du}{dt}\right) + f(t, u(t)) = 0$$

with boundary conditions. Solvability is equivalent to Urysohn equation:

$$u(t) + \int_0^1 K(s, t, u) f(s, u(s)) ds = 0, \quad (8.56)$$

where $K(s, t, u)$ is Green's function:

$$K(s, t, u) = \begin{cases} \int_0^t \frac{d\tau}{a(u(\tau))} \int_s^1 \frac{d\tau}{a(u(\tau))} & 0 \leq t \leq s \leq 1 \\ \int_0^s \frac{d\tau}{a(u(\tau))} \int_t^1 \frac{d\tau}{a(u(\tau))} & 0 \leq s \leq t \leq 1 \end{cases}$$

Computational Schemes for Solving Equations

Develop constructive methods for approximate solutions to:

$$u + K(u)Nu = 0$$

Three Main Approaches:

1. **Direct Iteration:** Mann iteration and variants
2. **Galerkin-type Methods:** Projection onto finite-dimensional subspaces
3. **Hybrid Approaches:** Combine iteration with projection

Computational schemes are particularly important when exact solutions are unavailable and we need convergent approximation sequences.

Mann Iteration and Variants

[Mann Iteration] Given initial $x_0 \in X$, define sequence $\{x_n\}$ by:

$$v_{n+1} = \left(\frac{n}{n+1}\right) v_n - \left(\frac{1}{n+1}\right) SN(S^* v_n + w)$$

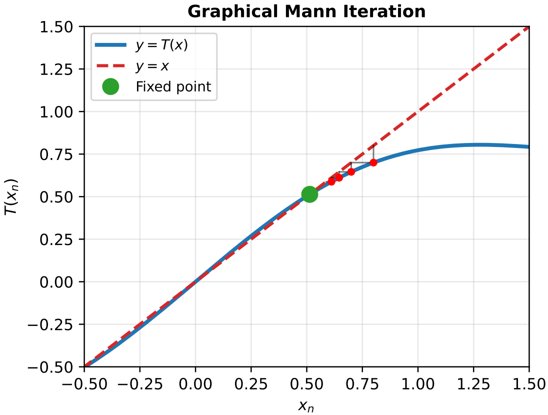
This converges to the unique solution of $Tv = 0$ with error estimate:

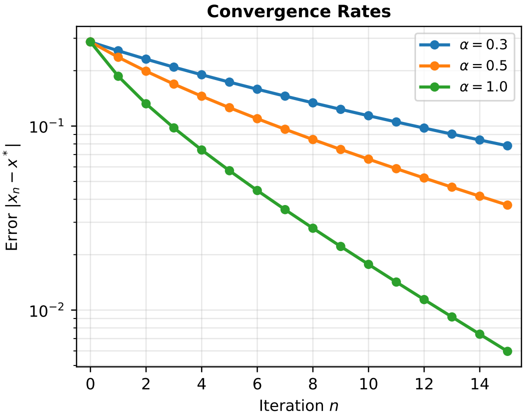
$$\|v_n - v\| = O(n^{-1/2})$$

Key Properties:

- Requires only one evaluation of nonlinear operator per step
- Linear convergence rate under suitable conditions
- Robust with respect to perturbations
- Works in reflexive Banach spaces

Mann Iteration Convergence





Left: Graphical visualization of Mann iteration steps. Fixed point is found by iterative refinement.
Right: Convergence rates for different step sizes.

Projection Methods: Galerkin Approach

Definition (Projection Scheme)

Let $\{X_n, P_n\}$ be a projection scheme in X, where:

- X_n are closed subspaces with $\dim(X_n) = n$
- $P_n : X \to X_n$ are continuous projections

The *Galerkin approximate equation* is:

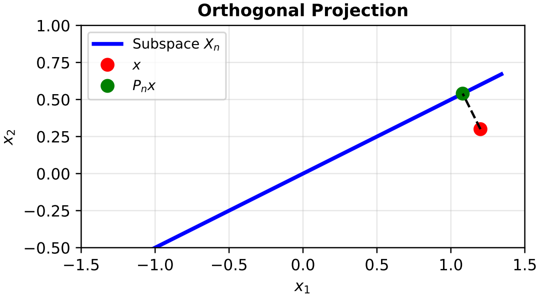
$$u_n + K_n(u_n)N_n u_n = 0, \quad (8.67)$$

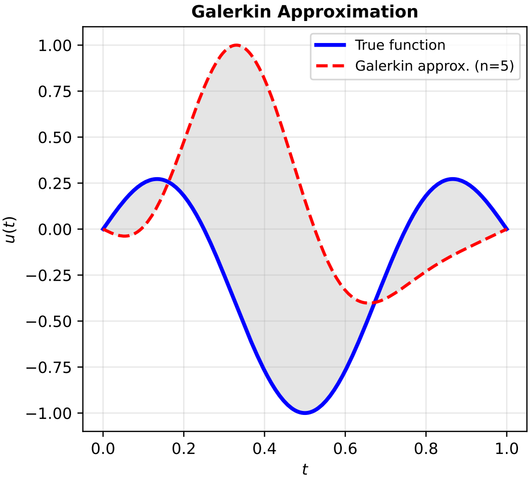
where $K_n(u) = P_n^* K(u) P_n$ and $N_n = P_n N P_n^*$.

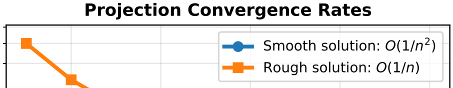
Definition (Approximate Solvability)

Equation (8.66) is *strongly (weakly) approximately solvable* if (8.67) has solution $u_n \in Y_n = (P_n^* X)$ such that $u_{n_k} \to u$ in X^* and u solves (8.66).

Projection Schemes: Properties and Convergence







Convergence of Projection Methods I

Theorem (Theorem 8.22)

Let X be a real reflexive Banach space with $\{X_n, P_n\}$ a projection scheme. For each $u \in C$ (closed bounded set in X^*), let $K(u) : X \to X^*$ be bounded linear operator and $N : X^* \to X$ continuous and bounded nonlinear.

Assume:

(a) $K(u)$ *is compact and monotone for each* $u \in C$

(b) $u_n \to u$ *in* C *implies* $K(u_n)v \to K(u)v$ *for all* $v \in X$

(c) $(u, Nu) > 0$ *for all* $u \in \partial C$

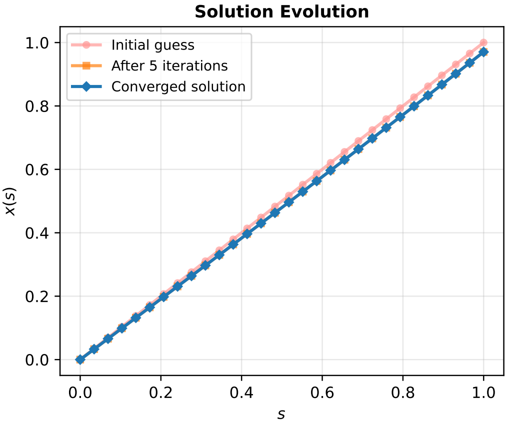
Then equation (8.66) is strongly approximately solvable. If solution is unique, the entire sequence $\{u_n\}$ converges to the unique solution.

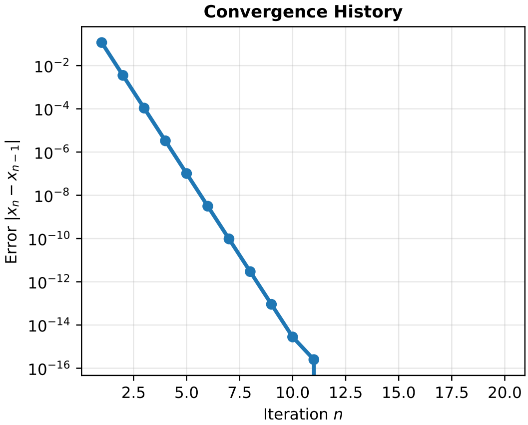
Convergence of Projection Methods II

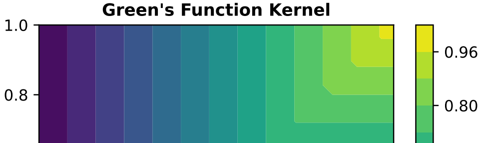
Proof Outline.

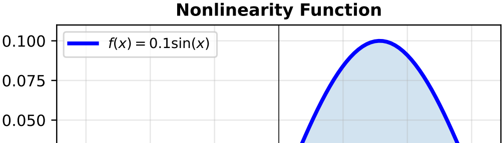
Define $T_n u = -K_n(u)N_n u$. Since T_n maps $C_n = C \cap Y_n$ into Y_n, and $T_n \neq \lambda u$ for $\lambda > 1$ and $u \in \partial C_n$ by Leray–Schauder principle, there exists u_n with $T_n u_n = u_n$. $\square$

Hammerstein Equation Example









Python Implementation: Simple Fixed Point

```
import numpy as np
from scipy.integrate import quad

def hammerstein_solver(kernel, nonlinearity, forcing, n_iter=20):
    """Solve Hammerstein equation via fixed-point iteration"""
    n_points = 30
    s = np.linspace(0, 1, n_points)
    ds = 1.0 / n_points

    x = s.copy()  # Initial guess
    errors = []

    for iteration in range(n_iter):
        # Compute integral term
        integral_term = np.zeros(n_points)
        for i in range(n_points):
            for j in range(n_points):
                integral_term[i] += (kernel(s[i], s[j]) *
                                    nonlinearity(x[j]) * ds)

        # Update: x(s) = g(s) - integral_term
        x_new = forcing(s) - integral_term

        # Track convergence
        error = np.linalg.norm(x_new - x)
        errors.append(error)
        x = x_new
```

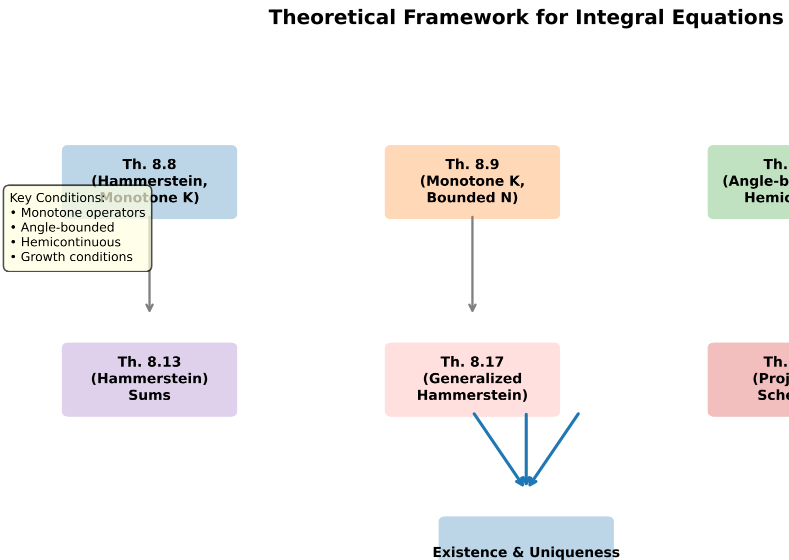
Comparative Analysis: Methods and Convergence

Method	**Type**	**Rate**	**Complexity**	**Robust**
Mann Iteration	Direct	Linear	Low	Yes
Galerkin (n=10)	Projection	$O(1/n^2)$	Medium	Yes
Petrov-Galerkin	Projection	$O(1/n^2)$	High	Moderate
Hybrid (Mann+Proj)	Combined	Super-linear	Medium	Yes

Selection Criteria:

- **Mann iteration**: When operator is inexpensive to evaluate
- **Galerkin**: When dimension reduction is beneficial
- **Hybrid**: For robustness and faster convergence

Theoretical Framework



Key Theorems Summary I

Existence and Uniqueness Results

- **Theorem 8.8**: Hammerstein with monotone K and hemicontinuous N
  - Condition: K strongly monotone
  - Conclusion: Unique solution with continuous dependence
- **Theorem 8.10**: Angle-bounded case
  - Condition: $k(1 + c^2)\|K\| < 1$
  - Conclusion: Unique solution with explicit estimate
- **Theorem 8.17**: Generalized Hammerstein
  - Condition: $K(u)$ angle-bounded with $\|K(u)\| \leq r$
  - Conclusion: At least one solution when $kr < (1 + c^2)$

Key Theorems Summary II

Computational Results

- **Theorem 8.22**: Projection methods
  - Strong approximate solvability when:
    1. $K(u)$ compact and monotone
    2. Continuity condition on K
    3. Boundary condition on ∂C
  - Entire sequence converges if solution is unique
  - $O(1/n^2)$ convergence for smooth solutions
- **Mann Iteration**: Direct approach
  - Simple to implement
  - Linear convergence rate
  - Works in general reflexive Banach spaces

Applications Beyond Theory

Classical Applications:

- Boundary value problems (ODE/PDE)
- Radiative transfer equations
- Potential theory
- Elasticity problems

Modern Applications:

- Neural networks (integral kernels)
- Machine learning (kernel methods)
- Signal processing
- Inverse problems

The theoretical framework developed in this chapter enables rigorous analysis of approximation methods for these diverse applications.

Connections to Previous Chapters

- **Chapter 5**: Fixed point theorems (Banach, Brouwer, Schauder)
  - Integral equations reduce to fixed point problems
  - Monotone operators studied here specialize those results
- **Chapter 4**: Monotone operators
  - Linear monotone operators form the basis
  - Maximal monotone extensions discussed
- **Chapters 2-3**: Iterative methods
  - Contraction mappings generalized to monotone setting
  - Mann iteration extends Picard iteration
- **Chapter 1**: Functional analysis foundations
  - Banach/Hilbert spaces structure used throughout
  - Weak convergence essential for projection methods

Open Problems and Future Directions

- **Computational Efficiency**
  - Optimal preconditioning for nonlinear systems
  - Combining with multigrid techniques
- **Nonsmooth Analysis**
  - Subdifferentials and proximal methods
  - Variational inequalities
- **Uncertainty Quantification**
  - Integral equations with random kernels
  - Convergence under noise
- **Scalability**
  - High-dimensional problems
  - Machine learning applications

Recommended Further Reading

- Pathak, R. P. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
- Krasnoselskii, M. A. (1964). *Topological Methods in the Theory of Nonlinear Integral Equations*.
- Zeidler, E. (1990). *Nonlinear Functional Analysis and its Applications*. Springer-Verlag.
- Lions, J. L. (1969). *Quelques méthodes de résolution des problèmes aux limites non linéaires*.

Key Takeaways

1. Integral equations (Hammerstein, Urysohn) are fundamental in applied mathematics
2. Monotone operator theory provides unified existence-uniqueness framework
3. Computational schemes (Mann iteration, projection methods) guarantee convergence
4. Hybrid approaches combine direct iteration with projection for robustness
5. Applications span classical analysis through modern machine learning

Final Thought

The interplay between abstract theory (monotone operators) and concrete methods (projection schemes) exemplifies how functional analysis enables both existence proofs and algorithmic solutions.